

Selected recipes for inverse problems

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Notations	
α	(lower case) scalar
α^*	complex conjugate of α
$\mathbf{x} \in \mathbb{C}^N$	(boldface lower case) complex vector
N	(upper case) cardinal (number of elements)
$\mathbf{H} \in \mathbb{C}^{N \times M}$	(boldface upper case) linear operator (matrix)
$f : \mathbb{C}^N \rightarrow \mathbb{C}^M$	(lower case) function
$L^2(\mathbb{R}^N)$	space of squared-integrable function
$\mathcal{H} : L^2(\mathbb{R}^N) \rightarrow L^2(\mathbb{R}^M)$	(calligraphic upper case) operator acting on functions

Basic properties	
$[\mathbf{H} \mathbf{x}]_m = \sum_n H_{m,n} x_n$	vector matrix product
$[\mathbf{H} \mathbf{G}]_{m,k} = \sum_n H_{m,n} G_{n,k}$	matrix product
$\mathbf{H}(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha \mathbf{H} \mathbf{x} + \beta \mathbf{H} \mathbf{y}$	linearity
$(\mathbf{x} \times \mathbf{y})_n = x_n y_n$	element wise product
$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$	identity matrix
$\mathbf{A} = \text{diag}(\mathbf{a}) = \begin{bmatrix} a_0 & 0 & 0 & \dots & 0 \\ 0 & a_1 & 0 & \dots & 0 \\ 0 & 0 & a_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_{N-1} \end{bmatrix}$	diagonal matrix

Adjoint, scalar product	
$(H^\dagger)_{i,j} = H_{j,i}^*$	adjoint (transpose conjugate)
$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n=0}^{N-1} x_n^* y_n = \mathbf{x}^\dagger \mathbf{y}$	dot (or scalar or inner) product
$\langle \mathbf{H} \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{H}^\dagger \mathbf{y} \rangle$	adjoint formal definition
$(\alpha \mathbf{H} + \beta \mathbf{G})^\dagger = \alpha^* \mathbf{H}^\dagger + \beta^* \mathbf{G}^\dagger$	
$(\mathbf{H} \mathbf{G})^\dagger = \mathbf{G}^\dagger \mathbf{H}^\dagger$	

Norms	
$\ \mathbf{x}\ _2^2 = \langle \mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^\dagger \mathbf{x} = \sum_{n=0}^{N-1} x_n ^2 = \sum_{n=0}^{N-1} x_n^* x_n$	ℓ^2 norm
$ \langle \mathbf{x}, \mathbf{y} \rangle \leq \ \mathbf{x}\ _2 \ \mathbf{y}\ _2$	(Cauchy-Schwartz)
$\ \mathbf{x} + \mathbf{y}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{y}\ _2^2 + 2 \langle \mathbf{x}, \mathbf{y} \rangle$	
$\ \mathbf{x}\ _p^p = \sum_{n=0}^{N-1} x_n ^p, \quad p \geq 1$	ℓ^p norm
$\ \mathbf{x} + \mathbf{y}\ _p \leq \ \mathbf{x}\ _p + \ \mathbf{y}\ _p$	triangular inequality
$\ \mathbf{H}\ _F^2 = \sum_{n,m} H_{i,j} ^2 = \text{tr}(\mathbf{H}^\dagger \mathbf{H})$	Frobenius norm

Inverse	
$\mathbf{H} \in \mathbb{C}^{N \times N}$ is called invertible (also nonsingular or nondegenerate) if there exists a matrix $\mathbf{G}$ such that:	
$\mathbf{H} \mathbf{G} = \mathbf{G} \mathbf{H} = \mathbf{I}$	
$\mathbf{G} = \mathbf{H}^{-1}$ is unique and is the inverse of $\mathbf{H}$	
$(\mathbf{H} \mathbf{M})^{-1} = \mathbf{M}^{-1} \mathbf{H}^{-1}$	for any $\mathbf{H}, \mathbf{M}$ invertible
$(\mathbf{H}^\dagger)^{-1} = (\mathbf{H}^{-1})^\dagger$	

Eigenvalue & eigenvectors of $\mathbf{H} \in \mathbb{C}^{N \times N}$	
$\lambda_i \in \mathbb{C}$ and $\mathbf{v}_i \in \mathbb{C}^N$ are the i^{th} eigenvalue and eigenvector respectively. They satisfy:	
$\mathbf{H} \mathbf{v}_i = \lambda_i \mathbf{v}_i$	
It leads to the eigendecomposition:	
$\mathbf{H} = \mathbf{Q} \text{diag}(\boldsymbol{\lambda}) \mathbf{Q}^{-1},$	
where the columns of $\mathbf{V}$ are the N eigenvectors of $\mathbf{H}$.	
■ $\boldsymbol{\lambda} = \text{eig}(\mathbf{H})$ is the spectrum of $\mathbf{H}$, ■ $C = \frac{\max \lambda }{\min \lambda }$ is the condition number, ■ $\text{rank}(\mathbf{H})$ is the number of non-zero element of $\boldsymbol{\lambda}$, ■ if $\lambda_i \neq 0, \forall i$, $\mathbf{H}$ is invertible and $\mathbf{H}^{-1} = \mathbf{Q} \text{diag}(\boldsymbol{\lambda})^{-1} \mathbf{Q}^{-1},$	

Peculiar matrices	
■ Unitary: $\mathbf{Q}^\dagger \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\dagger = \mathbf{I}$ ■ Hermitian: $\mathbf{H}^\dagger = \mathbf{H}$ and $\mathbf{H} = \mathbf{Q} \text{diag}(\boldsymbol{\lambda}) \mathbf{Q}^\dagger$ with $\boldsymbol{\lambda} \in \mathbb{R}^N$ and $\mathbf{Q}$ unitary ■ Positive semi-definite: $\lambda_i \geq 0, \forall i \iff \mathbf{x}^\dagger \mathbf{H} \mathbf{x} \geq 0, \forall \mathbf{x}$ ■ Toeplitz: $H_{i,j} = h_{i-j},$ ■ Circulant: Toeplitz with $H_{i,j} = h_{(i-j) \bmod N}$	

Trace and determinant	
■ trace: $\text{tr}(\mathbf{H}) = \sum_n H_{n,n} = \sum_n \lambda_n$	
$\text{tr}(\mathbf{H}^\dagger) = \text{tr}(\mathbf{H})^*$	$\text{tr}(\mathbf{H} \mathbf{G}) = \text{tr}(\mathbf{G} \mathbf{H})$
$\text{tr}(\alpha \mathbf{H} + \beta \mathbf{G}) = \alpha \text{tr}(\mathbf{H}) + \beta \text{tr}(\mathbf{G})$	
■ determinant: $\det(\mathbf{H}) = \prod_n \lambda_n$	
$\det(\mathbf{H}) \neq 0 \iff \mathbf{H}$ is invertible	$\det(\mathbf{H}^\dagger) = \det(\mathbf{H})^*$
$\det(\mathbf{H} \mathbf{G}) = \det(\mathbf{H}) \det(\mathbf{G})$	
$\det(\mathbf{H}^{-1}) = 1/\det(\mathbf{H})$	

Singular value decomposition	
For all matrices $\mathbf{H} \in \mathbb{C}^{N \times M}$ there exists two unitary matrices $\mathbf{U} \in \mathbb{C}^{N \times N}$ and $\mathbf{V} \in \mathbb{C}^{M \times M}$ and a real non-negative diagonal matrix $\boldsymbol{\Sigma} \in \mathbb{C}^{N \times M}$ (singular values: $\sigma_i = \Sigma_{i,i}, \forall i \leq \min(M, N)$) subject to:	
$\mathbf{H} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\dagger$	
■ columns of $\mathbf{U}$ are eigenvectors of $\mathbf{H} \mathbf{H}^\dagger,$ ■ columns of $\mathbf{V}$ are eigenvectors of $\mathbf{H}^\dagger \mathbf{H},$ ■ $\boldsymbol{\Sigma} = \sqrt{\text{diag}(\text{eig}(\mathbf{H} \mathbf{H}^\dagger))},$ ■ $\ \mathbf{H}\ _F^2 = \sum_i \sigma_i^2$ ■ $\text{rank}(\mathbf{H})$: number of non zero singular values,	

Inversion lemmas & Woodbury identity	
$\mathbf{B}^{-1} \mathbf{V} (\mathbf{A} - \mathbf{U} \mathbf{B}^{-1} \mathbf{V})^{-1} = (\mathbf{B} - \mathbf{V} \mathbf{A}^{-1} \mathbf{U})^{-1} \mathbf{V} \mathbf{A}^{-1}$	
$(\mathbf{A} - \mathbf{U} \mathbf{B}^{-1} \mathbf{V})^{-1} \mathbf{U} \mathbf{B}^{-1} = \mathbf{A}^{-1} \mathbf{U} (\mathbf{B} - \mathbf{V} \mathbf{A}^{-1} \mathbf{U})^{-1}$	
$(\mathbf{A} + \mathbf{U} \mathbf{B} \mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{U} (\mathbf{B}^{-1} + \mathbf{V} \mathbf{A}^{-1} \mathbf{U})^{-1} \mathbf{V} \mathbf{A}^{-1}$	

Moore-Penrose pseudo inverse	
The pseudo inverse of $\mathbf{H} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\dagger$ writes $\mathbf{H}^+ = \mathbf{V} \boldsymbol{\Sigma}^+ \mathbf{U}^\dagger$ with	
$\Sigma_{i,i}^+ = \begin{cases} \Sigma_{i,i}^{-1} & \text{if } \sigma_i \neq 0, \\ 0 & \text{otherwise.} \end{cases}$	
■ $\mathbf{H} \mathbf{H}^+ \mathbf{H} = \mathbf{H}$ and $\mathbf{H}^+ \mathbf{H} \mathbf{H}^+ = \mathbf{H}^+$ ■ $\mathbf{H}$ is square and $\text{rank}(\mathbf{H}) = N \Rightarrow \mathbf{H}^+ = \mathbf{H}^{-1}$ ■ $\mathbf{H}$ is broad: $\text{rank}(\mathbf{H}) \leq N$ and $\mathbf{H}^+ = \mathbf{H}^\dagger (\mathbf{H} \mathbf{H}^\dagger)^{-1}$ ■ $\mathbf{H}$ is tall: $\text{rank}(\mathbf{H}) \leq M$ and $\mathbf{H}^+ = (\mathbf{H}^\dagger \mathbf{H})^{-1} \mathbf{H}^\dagger$	

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Derivatives		
$J_{i,j} = \left[\frac{\partial \mathbf{x}}{\partial \mathbf{y}} \right]_{i,j} = \frac{\partial x_i}{\partial y_j}$	Jacobian matrix	
$[\nabla f]_i = \frac{\partial f}{\partial x_i}$	Gradient vector	
$[\nabla^2 f]_{i,j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$	Hessian matrix	

Derivation rules		
$\partial (\alpha \mathbf{A} + \beta \mathbf{B}) = \alpha \partial \mathbf{A} + \beta \partial \mathbf{B}$	linearity	
$\partial (\mathbf{A} \mathbf{B}) = (\partial \mathbf{A}) \mathbf{B} + \mathbf{A} (\partial \mathbf{B})$		
$\partial (\mathbf{A}^{-1}) = -\mathbf{A}^{-1} (\partial \mathbf{A}) \mathbf{A}^{-1}$		
$\partial (\mathbf{A}^\dagger) = (\partial \mathbf{A})^\dagger$		
$\frac{\partial \mathbf{x}^\dagger \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}^\dagger \mathbf{x}}{\partial \mathbf{x}} = \mathbf{y}$		
$\frac{\partial \mathbf{x}^\dagger \mathbf{A} \mathbf{x}}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\dagger) \mathbf{x}$		
$\frac{\partial f \circ g(\mathbf{x})}{\partial \mathbf{x}} = \sum_{m=0}^{M-1} \frac{\partial f}{\partial u_m} \frac{\partial u_m}{\partial \mathbf{x}} \Big _{\mathbf{u}=g(\mathbf{x})}$	chain rule	
$\frac{\partial (\mathbf{A} \mathbf{x} - \mathbf{y})^\dagger \mathbf{B} (\mathbf{A} \mathbf{x} - \mathbf{y})}{\partial \mathbf{A}} = (\mathbf{B} + \mathbf{B}^\dagger) (\mathbf{A} \mathbf{x} - \mathbf{y}) \mathbf{x}^\dagger$		

Continuous Fourier transform		
$\widehat{f}(\nu) = \mathcal{F}(f)(\nu) = \int_{-\infty}^{+\infty} f(t) \mathrm{e}^{-2\mathrm{i}\pi\nu t} \mathrm{d}t$	forward	
$f(t) = \mathcal{F}^{-1}\left(\widehat{f}\right)(t) = \int_{-\infty}^{+\infty} f(\nu) \mathrm{e}^{2\mathrm{i}\pi\nu t} \mathrm{d}\nu$	inverse	
$\mathcal{F}(\alpha f + \beta g) = \alpha \widehat{f} + \beta \widehat{g}$	linearity	
$\mathcal{F}(f(t-t_0)) = \mathrm{e}^{-2\mathrm{i}\pi\nu t_0} \widehat{f}(\nu)$	shift	
$\mathcal{F}(\mathrm{e}^{2\mathrm{i}\pi\nu_0 t} f(t)) = \widehat{f}(\nu - \nu_0)$	modulation	
$\mathcal{F}(f(at)) = \frac{1}{ a } \widehat{f}\left(\frac{\nu}{a}\right)$	scaling	
$\mathcal{F}(f^*)(\nu) = \widehat{f}^*(-\nu)$	conjugation	
$\int_{-\infty}^{+\infty} \left \widehat{f}(\nu) \right ^2 \mathrm{d}\nu = \int_{-\infty}^{+\infty} f(t) ^2 \mathrm{d}t$	Plancherel-Parseval	
$\widehat{\widehat{f}}(0) = \int_{-\infty}^{+\infty} f(t) \mathrm{d}t$	integration	
$\mathcal{F}(f^{(n)}) = (2\mathrm{i}\pi\nu)^n \widehat{f}(\nu)$	differentiation	
$\mathcal{F}(f * g) = \widehat{f} \widehat{g}$	convolution	
$\mathcal{F}(f \star f) = \left \widehat{f} \right ^2$	autocorrelation	

Discrete Fourier Transform		
$\mathbf{F} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)(N-1)} \end{bmatrix}$	with $\omega = \mathrm{e}^{-\frac{2\mathrm{i}\pi}{N}}$ the N^{th} root of unity.	
$\mathbf{F}^{-1} = \frac{1}{N} \mathbf{F}^\dagger$	orthogonality	
$\mathbf{U} = \frac{1}{\sqrt{N}} \mathbf{F}$	$\mathbf{U}$ is unitary	
$\ \mathbf{x}\ _2^2 = \frac{1}{N} \ \mathbf{F} \mathbf{x}\ _2^2$	Plancherel-Parseval	
$\ \mathbf{x}\ _1 \leq \ \mathbf{F} \mathbf{x}\ _1 \leq N \ \mathbf{x}\ _1$		

Circular convolution matrix $\mathbf{H}$		
$H_{0,i} = h_i$	impulse response (PSF)	
$\mathbf{H} = \mathbf{F}^{-1} \text{diag}(\widehat{\mathbf{h}}) \mathbf{F}$	diagonalization by Fourier	
$\widehat{\mathbf{h}} = \mathbf{F} \mathbf{h}$	eigenvalues spectrum	

Continuous probability distribution

$\mathbf{x} \in \mathbb{C}^N$ is a continuous random vector, it has a probability density function (pdf) $f_X(\mathbf{x})$ such that, for all $\mathbb{A} \subseteq \mathbb{C}^N$:

$$\Pr(\mathbf{x} \in \mathbb{A}) = \int_{\mathbb{A}} f_X(\mathbf{x}) \, \mathrm{d}\mathbf{x}$$

$\bar{\mathbf{x}} = \mathbb{E}(\mathbf{x}) = \langle \mathbf{x} \rangle = \int_{-\infty}^{+\infty} \mathbf{x} f_X(\mathbf{x}) \, \mathrm{d}\mathbf{x}$
Expectation (mean)

$\mathbb{E}(\alpha \mathbf{x} + \beta \mathbf{y} + \gamma) = \alpha \mathbb{E}(\mathbf{x}) + \beta \mathbb{E}(\mathbf{y}) + \gamma$
linearity

$\mathbf{C}_{\mathbf{x},\mathbf{y}} = \text{Cov}(\mathbf{x}, \mathbf{y}) = \mathbb{E}(\mathbf{x}\mathbf{y}^\dagger) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{y})^\dagger$
Cross-covariance

$\mathbf{C}_{\mathbf{x}} = \text{Cov}(\mathbf{x}, \mathbf{x}) = \mathbb{E}(\mathbf{x}\mathbf{x}^\dagger) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x})^\dagger$
Covariance

$\mathbf{C}_{\mathbf{x}}$ is an Hermitian matrix of size $N \times N$.

Independence & Uncorrelation		
$\Pr(X \mid Y) = \Pr(X)$	Independence	
$f_{X,Y}(\mathbf{x}, \mathbf{y}) = f_X(\mathbf{x}) f_Y(\mathbf{y})$	Independence	
$\text{Cov}(\alpha \mathbf{x} + \beta \mathbf{y} + \gamma) = \alpha^2 \text{Cov}(\mathbf{x}) + \beta^2 \text{Cov}(\mathbf{y})$	Uncorrelation	
$\text{Cov}(\mathbf{x}, \mathbf{y}) = 0$	Uncorrelation	
independence $\Rightarrow$ uncorrelation	uncorrelation $\nRightarrow$ independence	
$\Pr(X \mid Y) = \frac{\Pr(Y \mid X) \Pr(X)}{\Pr(Y)}$	Bayes rule	

Convexity		
$f : \mathbb{C}^N \rightarrow \mathbb{R}$ is strictly convex if and only if:		
$f(\lambda \mathbf{x} + (1-\lambda) \mathbf{y}) < \lambda f(\mathbf{x}) + (1-\lambda) f(\mathbf{y}) \quad \forall \mathbf{x}, \mathbf{y}, \lambda \in]0, 1[.$		
$\mathbf{x}^+ = \arg \min_{\mathbf{x}} f(\mathbf{x})$	global minimum	
$\mathbf{p} \in \partial f(\mathbf{x}) \Leftrightarrow f(\mathbf{x}) - f(\mathbf{x}') \geq \langle \mathbf{p}, \mathbf{x} - \mathbf{x}' \rangle, \forall \mathbf{x}'$	subgradient	

Gradient descent		
$f : \mathbb{C}^N \rightarrow \mathbb{R}$ is convex and differentiable with Lipschitz gradient L :		
$\ \nabla \mathbf{x} - \nabla \mathbf{y}\ _2 \leq L \ \mathbf{x} - \mathbf{y}\ _2$		
The following sequence converges toward a minimizer of f in $O(1/k)$:		
$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \gamma \nabla f(\mathbf{x}^{(k)})$ with $\gamma \in]0, 1/L]$		

Newton's method		
$f : \mathbb{C}^N \rightarrow \mathbb{R}$ is convex and twice differentiable, the sequence:		
$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \left(\nabla^2 f(\mathbf{x}^{(k)}) \right)^{-1} \nabla f(\mathbf{x}^{(k)})$		
converges toward a minimizer of f in $O(1/k^2)$:		

Projection		
$\mathbf{P} \in \mathbb{C}^{N \times N}$ is a projection on a subset $\mathbb{S} \subseteq \mathbb{C}^N$ and its indicator $\mathfrak{z}_{\mathbb{S}}$:		
$\mathbf{P} \mathbf{x} = \arg \min_{\mathbf{y}} \left(\mathfrak{z}_{\mathbb{S}}(\mathbf{y}) + \frac{1}{2} \ \mathbf{x} - \mathbf{y}\ ^2 \right) \quad \text{with } \mathfrak{z}_{\mathbb{S}}(\mathbf{x}) \begin{cases} 0 & \text{if } \mathbf{x} \in \mathbb{S}, \\ +\infty & \text{otherwise.} \end{cases}$		
$\mathbf{P} \mathbf{P} = \mathbf{P}.$	idempotent	
$\ \mathbf{P} \mathbf{x} - \mathbf{P} \mathbf{y}\ _2 \leq \ \mathbf{x} - \mathbf{y}\ _2$	$\mathbb{S}$ is convex $\Rightarrow \mathbf{P}$ is non-expansive	
$\mathbf{x}^{(k+1)} = \mathbf{P}(\mathbf{x}^{(k)} - \gamma \nabla f(\mathbf{x}^{(k)}))$	projected gradient descent	

Proximity operator		
f : a lower semi-continuous convex function, its proximity operator is:		
$\text{prox}_f(\mathbf{y}) = \arg \min_{\mathbf{y}} \left(f(\mathbf{y}) + \frac{1}{2} \ \mathbf{x} - \mathbf{y}\ ^2 \right)$		
$\mathbf{p} = \text{prox}_f(\mathbf{x}) \Leftrightarrow \mathbf{x} - \mathbf{p} \in \partial f(\mathbf{p})$ with $\partial f(\mathbf{p})$ the subgradient of f		
$\mathbf{x}^+ = \text{prox}_f(\mathbf{x}^+) \Leftrightarrow \mathbf{x}^+ = \arg \min_{\mathbf{x}} f(\mathbf{x})$		